MIL-P-50398 (AR)
AMENDMENT 5
17 November 1986
SUPERSEDING
AMENDMENT 4
10 January 1986

MILITARY SPECIFICATION

PROJECTILE, 165MM, TP
M624, METAL PARTS

This amendment forms a part of Military Specification MIL-P-50398 (AR) dated 18 May 1979, and is approved for use within the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1: Delete "MIL-I-6868 Inspection Process, Magnetic Particle."

PAGE 3

Add:

"3.1.1 Rotating band application. The rotating band shall be overlay welded using the gas metal arc welding process in accordance with the applicable drawings and specifications. Gas metal arc welding process in this case means the use of a consumable electrode with a non-electrode auxiliary wire. The width of the band shall be obtained by oscillating the welding head and the auxiliary feed system. The process for applying the rotating band, once established must be maintained, controlled, and monitored by the contractor. Procedures for maintaining, controlling and monitoring the process must be submitted to the technical agency for approval prior to start of production. Documented records of the process control must be made available to the Government upon request.

3.1.2 Foreign matter on band seat. Prior to banding, the band seat should be free from any contaminants that would be detrimental to the adhesion of the band to the band seat."

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3.5.1: Delete and substitute the following:

"3.5.1 Raw material. The billets, bars, tube or plate material from which projectile components are formed shall be free from cracks, splits, cold shuts, inclusions, pipe, porosity, folds, bursts, seams, laminations and other metal defects".

PAGE 4

3.6 d: Delete in its entirety.

3.8.2: Add to end of sentence: "(see 6.7)".

PAGE 6

Table I, Add:

Examination or Test column "Projectile Body (prior to painting) Band Chemical Analysis"
Number of samples - "1"
Requirement paragraph - "3.1.1"
Paragraph Reference/Inspection Method - "4.5.6.1"

PAGE 8

4.4.2.1.3, delete and substitute the following:

"4.4.2.1.3 Inspection of raw material prior to forming. Each slug, tube or disc obtained from bar, tube or plate stock shall be visually examined for surface indications of the types of metal defects described in 3.5.1 which are visible to the unaided eye on all exterior surfaces. Visual standards may be selected to aid in the determination of the slug's, tube's, or disc's acceptability".

PAGE 9

4.4.2.2, Major 101, Delete: "Runout...." and substitute: "True position....".

PAGE 10

4.4.2.2: Delete Minor 206 in its entirety.

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4.4.2.3: Add the following to the note block:

"A. See 4.4.3.1 for sample size."

PAGE 14

4.4.2.5 after line 124, add:

Major: "125; Band Chemical Analysis"

Under No. of Sample Units add "A"

Under requirement paragraph, add "3.1.1"

Under Notes, add "A". One body after banding from each four hours production. (See 4.4.3.5)."

PAGE 17

4.4.2.7, Title: Delete "Carton, Packing, Prior to Sealing" and substitute "Carton, Packed, Prior to Sealing (Base Plug)".

PAGE 18

4.4.2.8, Title: Delete "Cartons, Packed, After Sealing" and substitute "Carton, Packed, After Sealing (Base Plug)".

PAGE 19

4.4.2.9, Title: Delete "Pallet" and substitute "Pallet (Projectile Body)".

PAGE 20

4.4.3.1: Delete and substitute the following:

"4.4.3.1 Tensile strength of projectile body. Two projectile bodies from the first batch of a stress-relieved lot shall be subjected to this test. Following satisfactory compliance of the initial sample with the requirements of dwg. 10551955, two projectile bodies from every five stress relieved batches, or fraction thereof, of a stress-relieved lot shall be subjected to this test. Failure of any projectile body tensile specimen to meet the applicable requirements shall reject the lot of stress-relieved bodies which the specimen represents and shall require the establishment of a new stress-relieved lot."

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4.4.3.2: Delete and substitute the following:

"4.4.3.2 Tensile strength of base-plug. Two base-plugs from the first batch of a stress-relieved lot of base-plugs shall be subjected to this test. Following satisfactory compliance of the initial sample with the requirements of dwg. 10535957, two base-plugs from every five stress-relieved batches, or fraction thereof, of a stress-relieved lot shall be subjected to this test. Failure of any base-plug tensile specimen to meet the applicable requirements shall reject the lot of stress-relieved base-plugs which the specimen represents and shall require the establishment of a new stress-relieved lot".

4.4.3.3: Delete the second sentence and substitute the following:

"Failure to comply with the applicable specification and drawing requirements shall be cause for the rejection of the material represented by the failing samples."

Add:

"4.4.3.5 Band chemical analysis. One body after banding from each 4 hours production or fraction thereof for each welding machine in operation shall be subjected to this test. Failure to comply with the requirements of drawing shall be cause for rejection of the production represented by the sample."

PAGE 21

4.5.1.1: Delete and substitute the following:

"4.5.1.1 Tensile test of projectile body. From the samples provided in accordance with 4.4.3.1, two tensile test specimens conforming to the largest obtainable standard specimen for tubular products, as prescribed by ASTM E8, shall be taken from opposite sides of the body in the minimum wall thickness area rear of the ogive".

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4.5.2: Add the following sentence:

"The contractor shall prepare a written standard operating procedure, to include the inspection of projectile bodies and calibration procedures, which shall be submitted for the review and approval of the Contracting Officer prior to the submission of the first article sample (see 6.2)".

PAGE 22

4.5.2.1: Delete "(see 4.5.3.1)" from first sentence and delete the last sentence in its entirety.

4.5.3 Delete the third sentence and substitute:

"The body shall be mounted in an approved fixture in a manner such that at least fifty percent of the body from the nose end shall be submerged in water."

4.5.3: Add to the end of the paragraph:

"The contractor shall prepare a written standard operating procedure to include the inspection of projectile bodies and calibration procedures, which shall be submitted for the review and approval of the Contracting Officer prior to the submission of the first article sample (see 6.2)".

4.5.3.1, delete in its entirety.

* 4.5.5: Add to the end of first sentence "...with a minimum remaining tube life of 50%."

PAGE 23

* 4.5.5.1: First sentence, delete "...in a gun tube with a fifty percent remaining life minimum."

* 4.5.5.2: Delete from end of sentence "...in a gun tube with twenty-five percent remaining life, maximum."

4.5.5.4: Add to end of paragraph:

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"The proving ground shall prepare a written assembly procedure for the assembly of projectile metal parts ballistic samples and submit the procedure for review to the Contracting Officer prior to the assembly of the first article ballistic test samples (see 6.4)".

Add:

"4.5.6 Rotating band."

Add:

"4.5.6.1 Band chemical analysis. From the bodies selected in 4.4.3.5 sample chips shall be obtained from each band from the last .030 inches removed by machining prior to finish diameter. All chips shall be collected and identified with the shell from which the chips were machined. A minimum 5 gram sample from each complete specimen thus obtained from each band shall be dissolved and the analysis made from the individual solutions. The sample may not be cleaned with a magnet."

PAGE 26

Add:

"6.7 Burrs. Burrs that are loose, injurious to personnel, interfere with the assembly of other components, or interfere with inspection gaging are not permissible."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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(Project 1320-A899)

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